

Bayesian CUPED: Hierarchical Shrinkage for Variance Reduction in Online Experiments

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Abstract

CUPED (Controlled Experiments Using Pre-Experiment Data) is the dominant variance-reduction technique in online A/B testing. We establish a foundational result: when the pre-experiment covariate is discretised into K strata, **CUPED is a matching estimator**—it is algebraically equivalent to a precision-weighted post-stratification estimator that matches treated and control units within strata defined by the covariate. Formally, by the Frisch–Waugh–Lovell theorem, the CUPED coefficient is exactly a convex combination of within-stratum treatment effects (Theorem 1):

$$\hat{\beta}_{\text{CUPED}} = \sum_{k=1}^K \lambda_k \hat{\tau}_k, \quad \lambda_k \propto n_k p_k (1 - p_k),$$

where the weights are determined by stratum size and treatment balance—not by the population distribution of users. This variance-weighted matching targets the *variance-weighted average treatment effect* (VWATT) rather than the *population average treatment effect* (PATE), a distinction that is material whenever treatment effects are heterogeneous across users—the typical case in large-scale digital platforms.

We introduce *Bayesian CUPED*, a hierarchical generalisation that places a normal prior $\tau_k \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2)$ over stratum-level effects and estimates the PATE via partial pooling. Classical CUPED is recovered in the limit $\sigma_\tau^2 \rightarrow 0$ (Theorem 2), and the posterior mean shrinks each stratum estimate toward the global mean by a James–Stein factor $B_k = s_k^2 / (s_k^2 + \sigma_\tau^2)$ (Theorem 4).

We provide two implementations: a full MCMC estimator via PyMC with calibrated credible intervals, and a *closed-form Empirical Bayes* (EB) approximation requiring only $O(K)$ arithmetic operations while preserving the shrinkage properties of the hierarchical model. A simulation study across 21 regimes and $B = 300$ Monte Carlo replications shows that EB CUPED achieves the lowest mean RMSE (0.0743) of all five estimators evaluated, capturing approximately 61% of the efficiency gain of classical CUPED over the naive estimator while correctly targeting the PATE. In the most adversarial regime (heavy-tailed errors, low covariate signal, small samples, and sparse treatment), EB CUPED reduces RMSE by 16.4% over the naive estimator, whereas classical CUPED actually loses efficiency (RE = 0.99). Full simulation results are reported in the companion document (Farahat, 2026).

1 Introduction

Online A/B testing is the primary mechanism for measuring the causal impact of product changes at large-scale digital platforms. Treatment effect estimates often suffer from high variance, particularly when conversion events are rare or effect sizes are small. CUPED, introduced by [Deng et al. \(2013\)](#), reduces variance by incorporating a pre-experiment covariate:

$$Y_i = \alpha + \beta D_i + \gamma X_i + \epsilon_i, \quad (1)$$

where Y_i is the post-experiment outcome, D_i is the treatment indicator, and X_i is a pre-experiment metric (e.g., the user’s metric value in the week prior to the experiment). When X_i is predictive of Y_i , the regression adjustment absorbs a large share of outcome variance, increasing experiment sensitivity.

CUPED is now widely deployed across the industry and has been discussed in several subsequent papers ([Deng et al., 2013, 2017](#); [Poyarkov et al., 2016](#)). Yet a fundamental question has received little attention: *exactly what estimand does the CUPED coefficient target?* The answer, which we formalise in this paper, has practical consequences for how results are interpreted and acted upon.

Main contribution: CUPED is a matching estimator. Our central result (Theorem 1) shows that when X_i is discretised into K strata, the CUPED regression coefficient is *algebraically identical* to a precision-weighted post-stratification estimator. The Frisch–Waugh–Lovell (FWL) theorem reveals that CUPED residualises both the outcome and the treatment indicator on the stratum dummies, then regresses the residualised outcome on the residualised treatment. With discrete covariates, this residualisation is exactly *within-stratum demeaning*, which is precisely the within-group transformation used in matching estimators ([Angrist and Pischke, 2009](#)). The resulting weights $\lambda_k \propto n_k p_k (1 - p_k)$ match units on the stratum label and aggregate with precision-based weights, targeting the variance-weighted average treatment effect (VWATT) rather than the population average treatment effect (PATE).

This connects CUPED squarely to the matching literature. Just as nearest-neighbour matching matches each treated unit to its closest control and aggregates with frequency-based weights, CUPED matches units within covariate-defined strata and aggregates with variance-based weights. The distinction matters whenever treatment effects are heterogeneous: VWATT downweights strata with extreme treatment propensities, which may not coincide with the most policy-relevant strata.

Proposed remedy: Bayesian CUPED. We address the VWATT/PATE gap by placing a hierarchical prior over stratum effects. The resulting model implements James–Stein shrinkage across strata and explicitly targets the PATE via population weights $w_k = n_k/n$. We show this subsumes classical CUPED and all variants of stratified estimation as limiting cases, and we provide a closed-form Empirical Bayes approximation that is deployable as a drop-in replacement for OLS CUPED with no MCMC and no infrastructure changes.

Paper outline. Section 2 derives the matching interpretation of CUPED and formalises the VWATT/PATE gap. Section 3 introduces the Bayesian hierarchical model. Section 4 derives the closed-form Empirical Bayes estimator and provides the algorithm. Section 5 presents the simulation evidence. Section 6 concludes with practical guidance. Complete simulation results across all 21 regimes and all six evaluation metrics are available in the companion document (Farahat, 2026).

2 CUPED as a Matching Estimator

2.1 The Frisch–Waugh–Lovell Theorem

The Frisch–Waugh–Lovell (FWL) theorem (Frisch and Waugh, 1933; Lovell, 1963) states that in the partitioned regression

$$\mathbf{Y} = \mathbf{D}\beta + \mathbf{X}\gamma + \epsilon,$$

the OLS coefficient on $\mathbf{D}$ equals

$$\hat{\beta} = (\mathbf{D}'\mathbf{M}_X\mathbf{D})^{-1}\mathbf{D}'\mathbf{M}_X\mathbf{Y}, \quad (2)$$

where $\mathbf{M}_X = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the annihilator matrix that projects onto the space orthogonal to $\mathbf{X}$. The estimator uses only the variation in $\mathbf{D}$ that is orthogonal to the covariates—exactly the variation that identifies causal effects.

2.2 Within-Stratum Demeaning

When $\mathbf{X}$ consists of stratum indicator dummies ($X_{ik} = 1$ if unit i is in stratum k), the annihilator matrix performs *within-stratum demeaning*:

$$\tilde{D}_i = (\mathbf{M}_X\mathbf{D})_i = D_i - p_k, \quad p_k = n_k^T/n_k, \quad (3)$$

$$\tilde{Y}_i = (\mathbf{M}_X\mathbf{Y})_i = Y_i - \bar{Y}_k, \quad (4)$$

for all i in stratum k , where p_k is the stratum treatment propensity and $\bar{Y}_k$ is the stratum mean outcome. As Angrist and Pischke (2009, pp. 74–75) emphasise, regression on a full set of dummies subtracts the relevant group mean—the same within-group transformation used in panel data models and matching estimators.

The FWL residual $\tilde{D}_i$ captures how much unit i ’s treatment status deviates from its stratum’s treatment rate. All *between-stratum* variation in treatment assignment has been removed; only within-stratum variation is used to identify the effect. This is precisely the matching principle: compare treated and control units that are alike on observed covariates.

2.3 CUPED as a Weighted Sum of Within-Stratum Effects

Theorem 1 (CUPED as a Matching Estimator). *Let X_i be discretised into K strata with n_k units in stratum k and treatment propensity $p_k = n_k^T/n_k$. Let $\hat{\tau}_k = \bar{Y}_k^T - \bar{Y}_k^C$ be the within-stratum difference-in-means. The OLS estimator in regression (1) with stratum dummies for*

X_i satisfies:

$$\hat{\beta}_{CUPED} = \sum_{k=1}^K \lambda_k \hat{\tau}_k, \quad \lambda_k = \frac{n_k p_k (1 - p_k)}{\sum_{j=1}^K n_j p_j (1 - p_j)}. \quad (5)$$

The weights satisfy $\lambda_k \geq 0$ and $\sum_k \lambda_k = 1$.

Proof. By FWL, $\hat{\beta} = (\tilde{\mathbf{D}}' \tilde{\mathbf{Y}}) / (\tilde{\mathbf{D}}' \tilde{\mathbf{D}})$. For units in stratum k :

$$\sum_{i \in k} \tilde{D}_i \tilde{Y}_i = \sum_{i \in k} (D_i - p_k)(Y_i - \bar{Y}_k) = n_k p_k (1 - p_k) \hat{\tau}_k, \quad (6)$$

$$\sum_{i \in k} \tilde{D}_i^2 = \sum_{i \in k} (D_i - p_k)^2 = n_k p_k (1 - p_k). \quad (7)$$

Summing over strata and dividing gives (5). A complete step-by-step derivation is in Appendix 6. $\square$

Connection to matching. The representation (5) shows that CUPED is a weighted aggregation of within-stratum comparisons, with weights proportional to the *conditional variance of treatment* in each stratum. Angrist and Pischke (2009, pp. 70–80) develop precisely this connection: the OLS coefficient in a model with covariates can be written as the average of conditional covariances divided by the average conditional variance—a variance-weighted average of stratum-specific treatment effects. In our notation, $\lambda_k \propto n_k p_k (1 - p_k)$ is the conditional variance of D_i in stratum k , scaled by stratum size.

Table 1 shows the forces that determine each stratum’s weight.

Table 1: Determinants of stratum weights λ_k in CUPED.

Factor	Direction	Intuition
Stratum size n_k	$\uparrow \Rightarrow \lambda_k \uparrow$	More observations $\Rightarrow$ more precise comparison
Balanced treatment $p_k \rightarrow 0.5$	$\uparrow \Rightarrow \lambda_k \uparrow$	Variance of D is maximised at $p_k = 0.5$
Sparse/dense treatment $p_k \rightarrow 0, 1$	$\downarrow \Rightarrow \lambda_k \downarrow$	Little within-stratum variation in D

2.4 VWATT versus PATE

Theorem 1 reveals that CUPED’s population estimand is the *variance-weighted average treatment effect*:

$$\beta_{VWATT} = \frac{\sum_k n_k p_k (1 - p_k) \tau_k}{\sum_k n_k p_k (1 - p_k)}, \quad (8)$$

while the quantity of interest for business decisions is typically the *population average treatment effect*:

$$\tau_{PATE} = \sum_k \frac{n_k}{n} \tau_k. \quad (9)$$

When effects are constant ($\tau_k = \tau$ for all k), $\beta_{VWATT} = \tau_{PATE}$. When effects are heterogeneous—the typical case in large-scale digital platforms, where high-engagement users

respond differently from occasional users—the two diverge. Strata with strongly imbalanced treatment assignment (e.g., $p_k = 0.05$ in a sparse-treatment design) receive near-zero weight in β_{VWATT} regardless of their population share n_k/n . If treatment effects are large precisely in those strata, CUPED will understate the average effect that would be observed if the treatment were rolled out.

This distinction echoes the discussion in Angrist and Pischke (2009, pp. 80–82) on the interpretation of regression coefficients under heterogeneous treatment effects: the researcher must decide whether the variance-weighted average is the object of interest. For platform experiments where the goal is to estimate the effect of a full product launch, the PATE is the correct estimand.

3 Bayesian Hierarchical CUPED

3.1 Motivation: Partial Pooling Across Strata

With K strata, OLS CUPED estimates each stratum’s contribution independently and then aggregates with fixed precision weights. When some strata are small, the within-stratum estimates $\hat{\tau}_k$ are noisy, and even a perfect covariate cannot rescue the aggregate. A hierarchical model addresses this by allowing information to flow across strata: a stratum with a noisy estimate borrows strength from adjacent strata, shrinking toward the global mean in proportion to its own unreliability.

3.2 Hierarchical Model Specification

$$\text{Likelihood: } Y_{ik} \sim \mathcal{N}(\alpha_k + \tau_k D_{ik}, \sigma_y^2), \quad i = 1, \dots, n_k, \quad k = 1, \dots, K, \quad (10)$$

$$\text{Treatment effects: } \tau_k \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \quad (11)$$

$$\text{Hyperpriors: } \alpha_k \sim \mathcal{N}(0, 2), \quad \mu_\tau \sim \mathcal{N}(0, 1), \quad \sigma_\tau, \sigma_y \sim \text{Exponential}(1). \quad (12)$$

The prior $\tau_k \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2)$ encodes the belief that stratum effects are drawn from a common distribution. The parameter σ_τ governs heterogeneity: when σ_τ is large, the strata are nearly independent; when it is small, they share a common effect. The data determine σ_τ through the posterior. The population ATE is the derived quantity:

$$\tau_{\text{pop}} = \sum_{k=1}^K \frac{n_k}{n} \tau_k. \quad (13)$$

3.3 Classical CUPED as a Limiting Case

Theorem 2 (CUPED as a Limit of Bayesian CUPED). *Under the hierarchical model (10)–(12), as $\sigma_\tau^2 \rightarrow 0$ the model collapses to a homogeneous-effect specification $\tau_k = \mu_\tau$ for all k , and the estimator reduces to classical CUPED regression with a single treatment coefficient.*

Corollary 3. *Three regimes arise:*

- $\sigma_\tau = 0$: complete pooling; reduces to classical OLS CUPED.

- $0 < \sigma_\tau < \infty$: partial pooling; the operative Bayesian CUPED regime.
- $\sigma_\tau \rightarrow \infty$: no pooling; equivalent to fully stratified OLS.

Bayesian CUPED provides a continuous spectrum between all existing stratification-based estimators, with the data determining the degree of shrinkage.

4 Empirical Bayes CUPED: A Closed-Form Estimator

4.1 Setup and Posterior

Full MCMC inference is accurate but may be computationally demanding for production systems running hundreds of simultaneous experiments. We derive a closed-form Empirical Bayes (EB) approximation that preserves the shrinkage properties while requiring only $O(K)$ arithmetic operations.

Let $\hat{\tau}_k = \bar{Y}_k^T - \bar{Y}_k^C$ be the within-stratum difference-in-means with sampling variance $s_k^2 = \hat{\sigma}_{k,T}^2/n_k^T + \hat{\sigma}_{k,C}^2/n_k^C$. Under the hierarchical model, by normal-normal conjugacy:

$$\tau_k \mid \hat{\tau}_k \sim \mathcal{N}(m_k, v_k), \quad v_k = \left(\frac{1}{s_k^2} + \frac{1}{\sigma_\tau^2} \right)^{-1}, \quad m_k = v_k \left(\frac{\hat{\tau}_k}{s_k^2} + \frac{\mu_\tau}{\sigma_\tau^2} \right). \quad (14)$$

Theorem 4 (Empirical Bayes Shrinkage). *The posterior mean treatment effect in stratum k is:*

$$\mathbb{E}[\tau_k \mid \hat{\tau}_k] = (1 - B_k)\hat{\tau}_k + B_k \mu_\tau, \quad B_k = \frac{s_k^2}{s_k^2 + \sigma_\tau^2}. \quad (15)$$

The shrinkage factor $B_k \in [0, 1]$ measures the fraction of stratum k 's apparent variance attributable to sampling noise rather than true effect heterogeneity.

When $s_k^2 \gg \sigma_\tau^2$ (stratum is noisy), $B_k \rightarrow 1$ and the estimate is pulled toward the global mean. When $s_k^2 \ll \sigma_\tau^2$ (stratum has a precise, distinctive effect), $B_k \rightarrow 0$ and the raw estimate is retained. This is the James–Stein mechanism (James and Stein, 1961) made adaptive to the signal-to-noise ratio in each stratum.

4.2 Hyperparameter Estimation

In the EB approach, μ_τ and σ_τ^2 are estimated from the observed $\hat{\tau}_k$:

$$\hat{\mu}_\tau = \frac{\sum_k s_k^{-2} \hat{\tau}_k}{\sum_k s_k^{-2}}, \quad (16)$$

$$\hat{\sigma}_\tau^2 = \max \left(0, \frac{1}{K-1} \sum_k (\hat{\tau}_k - \hat{\mu}_\tau)^2 - \frac{1}{K} \sum_k s_k^2 \right). \quad (17)$$

The precision-weighted mean $\hat{\mu}_\tau$ gives more weight to strata whose estimates are more reliable. The method-of-moments estimator $\hat{\sigma}_\tau^2$ subtracts the average sampling variance to recover only the between-stratum signal; the max operator enforces non-negativity, which corresponds to complete pooling when the observed spread is consistent with sampling noise alone.

4.3 The EB CUPED Estimator

With estimated hyperparameters, the shrinkage estimates are:

$$\hat{\tau}_k^{\text{EB}} = (1 - \hat{B}_k)\hat{\tau}_k + \hat{B}_k \hat{\mu}_\tau, \quad \hat{B}_k = \frac{s_k^2}{s_k^2 + \hat{\sigma}_\tau^2}. \quad (18)$$

The population ATE estimate is:

$$\hat{\tau}_{\text{EB}} = \sum_{k=1}^K \frac{n_k}{n} \hat{\tau}_k^{\text{EB}}. \quad (19)$$

Note the contrast with OLS CUPED: EB CUPED uses population weights n_k/n to target the PATE, whereas OLS CUPED uses precision weights $\lambda_k \propto n_k p_k(1 - p_k)$ and targets the VWATT.

Algorithm 1 encapsulates the complete procedure.

Algorithm 1: Empirical Bayes CUPED (closed-form)

Input: Outcomes $\{Y_i\}$, treatments $\{D_i\}$, stratum indices $\{s(i)\}$

Output: Population ATE $\hat{\tau}^{\text{EB}}$, stratum ATEs $\{\hat{\tau}_k^{\text{EB}}\}$, shrinkage factors $\{B_k\}$

for $k = 1, \dots, K$ **do**

$\hat{\tau}_k \leftarrow \bar{Y}_k^T - \bar{Y}_k^C;$
 $s_k^2 \leftarrow \hat{\sigma}_{k,T}^2/n_k^T + \hat{\sigma}_{k,C}^2/n_k^C;$

$\hat{\mu}_\tau \leftarrow (\sum_k s_k^{-2} \hat{\tau}_k) / (\sum_k s_k^{-2});$ // Precision-weighted global mean

$\hat{\sigma}_\tau^2 \leftarrow \max(0, \frac{1}{K-1} \sum_k (\hat{\tau}_k - \hat{\mu}_\tau)^2 - \frac{1}{K} \sum_k s_k^2);$ // Method-of-moments

 heterogeneity

for $k = 1, \dots, K$ **do**

$B_k \leftarrow s_k^2 / (s_k^2 + \hat{\sigma}_\tau^2);$
 $\hat{\tau}_k^{\text{EB}} \leftarrow (1 - B_k)\hat{\tau}_k + B_k \hat{\mu}_\tau;$

$\hat{\tau}^{\text{EB}} \leftarrow \sum_k (n_k/n) \hat{\tau}_k^{\text{EB}};$

return $\hat{\tau}^{\text{EB}}, \{\hat{\tau}_k^{\text{EB}}\}, \{B_k\}$

Practical advantages. EB CUPED requires no MCMC, no distributional assumptions beyond normality of $\hat{\tau}_k$, and can be implemented in six lines of code as a post-processing step on any existing stratified analysis output. The shrinkage factor B_k is a free diagnostic: strata with $B_k > 0.8$ are too noisy for reliable inference and should prompt experiment extension or stratum consolidation.

5 Simulation Study

5.1 Design

We evaluate five estimators:

E1. Naive DIM: difference in means, no covariate adjustment.

- E2. CUPED OLS:** standard CUPED (Deng et al., 2013); targets VWATT.
- E3. Stratified OLS:** OLS with stratum fixed effects.
- E4. EB CUPED** (proposed, closed-form): Algorithm 1; targets PATE.
- E5. Bayesian MCMC CUPED** (proposed, full posterior): hierarchical model via PyMC with 4 chains, 2,000 draws.

The data-generating process is

$$Y_i = \rho X_i + \tau_{s(i)} D_i + \varepsilon_i, \quad (20)$$

where $X_i \sim \text{Gamma}(2, 1)$ is discretised into K quantile strata, $D_i \sim \text{Bernoulli}(p_{\text{treat}})$, and true stratum effects are $\tau_k \in \{0.05, 0.10, 0.20, 0.30, 0.45\}$ (heterogeneous) or $\tau_k = 0.20$ (homogeneous). The error ε_i is drawn from $\mathcal{N}(0, \sigma^2)$ (baseline), t_3 (heavy-tail), or a centred LogNormal(0, 0.5) (skewed). We run $B = 300$ Monte Carlo replications across 21 regimes that individually vary autocorrelation ρ , error distribution, effect heterogeneity, sample size n , number of strata K , treatment balance p_{treat} , and noise scale σ . Two compound regimes (**worst_case** and **best_case**) stress-test all estimators simultaneously. The true estimand is the population-weighted ATE $\tau^* = \sum_k (n_k/n) \tau_k$. Full regime definitions are in the companion document (Farahat, 2026).

5.2 Key Results: RMSE

Table 2 reports RMSE for the four fast estimators across all 21 regimes ($B = 300$ replications). The bold entry in each row is the best-performing estimator; an asterisk (*) marks regimes where EB CUPED strictly beats OLS CUPED.

Averaged across all 21 regimes, EB CUPED achieves the *lowest* mean RMSE (0.0743), marginally below OLS CUPED (0.0754) and Stratified OLS (0.0756), and substantially below Naive (0.0889). The key story, however, is in the tails:

- **Heavy-tailed errors (heavy_tail).** EB CUPED (0.0771) is the only estimator that improves over both OLS variants (≈ 0.0796), reducing RMSE by 3.3%. Heavy tails inflate s_k^2 , driving $B_k \rightarrow 1$ and pulling noisy stratum estimates toward the global mean—an adaptive mechanism unavailable to OLS.
- **Worst case.** Combining heavy tails, low autocorrelation ($\rho = 0.1$), small samples ($n = 300$), sparse treatment ($p = 0.20$), and high noise ($\sigma = 2$), OLS CUPED (0.5363) is *worse* than Naive (0.5327): without covariate signal, the regression adjustment wastes degrees of freedom. EB CUPED (0.4456) achieves a 16.4% improvement over Naive *purely* from cross-stratum pooling, which remains effective even when the covariate is uninformative.

5.3 Key Results: Bias

Table 3 reports Monte Carlo bias for the four estimators.

All estimators have negligible bias in well-specified settings ($|\text{bias}| \leq 0.011$ for OLS variants across all 21 regimes). In the worst case, EB CUPED (bias = 0.013) achieves the smallest absolute bias, 68% lower than OLS CUPED (0.040), because the shrinkage corrects

Table 2: Root Mean Squared Error across 21 simulation regimes ($B = 300$). **Bold** = best in row. * = EB CUPED improves over OLS CUPED.

Regime	E1 Naive	E2 CUPED OLS	E3 Strat. OLS	E4 EB CUPED
baseline	0.0538	0.0437	0.0436	0.0463
high_autocorr	0.0694	0.0437	0.0436	0.0507
low_autocorr	0.0445	0.0437	0.0436	0.0441
heavy_tail*	0.0861	0.0797	0.0795	0.0771
skewed	0.0400	0.0256	0.0256	0.0298
homogeneous	0.0522	0.0436	0.0436	0.0451
heterogeneous	0.0611	0.0437	0.0436	0.0483
large_effects	0.0573	0.0438	0.0436	0.0460
small_effects	0.0525	0.0436	0.0436	0.0453
small_sample	0.1519	0.1179	0.1187	0.1223
medium_sample	0.0766	0.0592	0.0591	0.0620
large_sample	0.0233	0.0192	0.0192	0.0202
few_strata	0.0539	0.0435	0.0437	0.0482
many_strata	0.0536	0.0436	0.0439	0.0452
balanced	0.0538	0.0437	0.0436	0.0463
imbalanced	0.0699	0.0590	0.0589	0.0631
sparse_treatment	0.1229	0.0976	0.0974	0.1004
low_noise	0.0338	0.0132	0.0131	0.0199
high_noise	0.1349	0.1309	0.1307	0.1329
worst_case*	0.5327	0.5363	0.5411	0.4456
best_case	0.0425	0.0080	0.0080	0.0208
Mean	0.0889	0.0754	0.0756	0.0743

Table 3: Monte Carlo bias $B^{-1} \sum_b (\hat{\tau}^{(b)} - \tau^*)$ across selected regimes ($B = 300$).

Regime	E1 Naive	E2 CUPED OLS	E3 Strat. OLS	E4 EB CUPED
baseline	-0.0035	-0.0010	-0.0009	-0.0066
high_autocorr	-0.0054	-0.0010	-0.0009	-0.0158
heavy_tail	-0.0033	-0.0008	-0.0006	-0.0019
homogeneous	-0.0032	-0.0009	-0.0009	-0.0005
small_sample	0.0149	0.0066	0.0058	0.0004
imbalanced	-0.0101	-0.0072	-0.0070	-0.0154
worst_case	0.0404	0.0402	0.0391	0.0131
best_case	-0.0005	-0.0003	-0.0004	-0.0143
Mean bias	0.0054	0.0038	0.0036	0.0062

for the inflated stratum-level noise that biases OLS in small samples. EB CUPED exhibits a systematic negative bias under high autocorrelation (-0.016) and strong heterogeneity (-0.011) due to mild over-shrinkage when the moment estimator $\hat{\sigma}_\tau^2$ underestimates heterogeneity; this bias is an order of magnitude smaller than the corresponding RMSE gain. Complete bias, variance, MAE, relative efficiency, and coverage results for all 21 regimes are reported in the companion simulation document (Farahat, 2026).

5.4 Bayesian MCMC: Targeted Comparison

Full MCMC is evaluated across five representative regimes with $B = 30$ replications (Table 4).

Table 4: RMSE including Bayesian MCMC (E5), $B = 30$ replications.

Regime	E1 Naive	E2 CUPED OLS	E3 Strat. OLS	E4 EB CUPED	E5 MCMC
baseline	0.0541	0.0439	0.0438	0.0461	0.0421
heavy_tail	0.0873	0.0809	0.0806	0.0779	0.0791
homogeneous	0.0518	0.0435	0.0432	0.0448	0.0440
small_sample	0.1531	0.1183	0.1192	0.1214	0.1106
worst_case	0.5441	0.5512	0.5558	0.4501	0.4238

MCMC achieves the lowest RMSE in 3 of 5 regimes. Its largest advantage is at $n = 300$, where MCMC (0.111) improves over EB CUPED (0.121) by 8.9% and over OLS CUPED (0.118) by 6.4%, reflecting the benefit of marginalising over hyperparameter uncertainty when data are scarce. MCMC credible interval coverage is near-nominal in all five tested regimes (range 0.93–0.97), confirming proper propagation of hyperparameter uncertainty.

6 Conclusion

This paper has shown that **CUPED is a matching estimator**: when the pre-experiment covariate is discretised, the CUPED coefficient is exactly a precision-weighted post-stratification estimator that matches treated and control units within covariate strata. The matching weights $\lambda_k \propto n_k p_k (1 - p_k)$ target the VWATTrather than the PATE, a distinction that is material whenever treatment effects are heterogeneous across users—the typical setting in large-scale online experiments.

Bayesian CUPED corrects this by design: the hierarchical model targets the PATE, implements adaptive James–Stein shrinkage across strata, nests classical CUPED as a limiting case, and provides a continuous spectrum between complete pooling and no-pooling stratified estimation. The Empirical Bayes approximation makes this correction practically free: six arithmetic steps, no MCMC, no infrastructure changes, and a free diagnostic (B_k) for identifying strata too noisy for reliable inference.

Summary of contributions.

1. **Theorem 1** (CUPED as matching estimator): the CUPED coefficient equals a convex combination of within-stratum ATEs with variance-based matching weights, connecting CUPED to the regression-as-matching framework of Angrist and Pischke (2009).
2. **Theorem 2 and Corollary 3**: classical CUPED is the $\sigma_\tau \rightarrow 0$ limit of Bayesian CUPED; the hierarchical model spans all existing stratification estimators.
3. **Theorem 4** (EB shrinkage): the closed-form James–Stein estimator achieves the lowest mean RMSE across 21 simulation regimes while correctly targeting the PATE.
4. **Simulation evidence**: EB CUPED uniquely improves over Naive in the worst-case regime where OLS CUPED fails; MCMC provides the best small-sample performance with calibrated uncertainty quantification.

Practical guidance.

- R1. EB CUPED** is the recommended default for large-scale production systems ($n \gg K$): it targets the PATE, is efficient, and provides shrinkage diagnostics.
- R2. Bayesian MCMC** is preferred when $n < 1,000$ or when calibrated uncertainty quantification is required.
- R3. OLS CUPED** remains the most efficient option when covariate signal is extremely strong ($\rho \approx 0.95$) and the VWATT/PATE distinction is immaterial.

For the platforms where CUPED was born, the Empirical Bayes estimator represents a principled, deployable upgrade to the core sensitivity machinery of online experimentation—one that correctly targets the business-relevant estimand and is more robust in the adversarial conditions that regularly arise in practice.

Appendix A: Complete Derivation of Theorem 1

Starting from the FWL representation $\hat{\beta} = \tilde{\mathbf{D}}'\tilde{\mathbf{Y}}/\tilde{\mathbf{D}}'\tilde{\mathbf{D}}$ with $\tilde{D}_i = D_i - p_k$ and $\tilde{Y}_i = Y_i - \bar{Y}_k$ for $i \in k$:

Numerator.

$$\sum_{i \in k} \tilde{D}_i \tilde{Y}_i = \sum_{i \in k} (D_i - p_k)(Y_i - \bar{Y}_k) \quad (21)$$

$$= n_k^T \bar{Y}_k^T - p_k n_k \bar{Y}_k - \bar{Y}_k n_k^T + p_k \bar{Y}_k n_k \quad (22)$$

$$= n_k^T (\bar{Y}_k^T - \bar{Y}_k) = n_k p_k (1 - p_k) \hat{\tau}_k. \quad (23)$$

Denominator.

$$\sum_{i \in k} \tilde{D}_i^2 = n_k^T (1 - p_k)^2 + n_k^C p_k^2 = n_k p_k (1 - p_k). \quad (24)$$

Assembly.

$$\hat{\beta} = \frac{\sum_k n_k p_k (1 - p_k) \hat{\tau}_k}{\sum_k n_k p_k (1 - p_k)} = \sum_k \lambda_k \hat{\tau}_k. \quad \square \quad (25)$$

Appendix B: MCMC Implementation in PyMC

```
import pymc as pm, numpy as np

# stratum_idx: integer array (0..K-1), treat: 0/1 array, y: outcome array
K = len(np.unique(stratum_idx))
strata_shares = np.array([(stratum_idx == k).mean() for k in range(K)])

with pm.Model() as cuped_model:
    mu_tau = pm.Normal('mu_tau', mu=0, sigma=1)
    sigma_tau = pm.Exponential('sigma_tau', lam=1)
    alpha_k = pm.Normal('alpha_k', mu=0, sigma=2, shape=K)
    tau_k = pm.Normal('tau_k', mu=mu_tau, sigma=sigma_tau, shape=K)
    sigma_y = pm.Exponential('sigma_y', lam=1)

    mu_y = alpha_k[stratum_idx] + tau_k[stratum_idx] * treat
    pm.Normal('y_obs', mu=mu_y, sigma=sigma_y, observed=y)

    pm.Deterministic('pop_ate', pm.math.dot(strata_shares, tau_k))
    pm.Deterministic('shrinkage', sigma_y**2 / (sigma_y**2 + sigma_tau**2))

    trace = pm.sample(draws=2000, tune=1000, chains=4, target_accept=0.9)
```

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